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AI-powered non-destructive testing for smart manufacturing of carbon-negative biopolymer-bound soil composite



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Biopolymer-bound soil composite (BSC) is a materials technology that uses nature-based polymers to form carbon-negative building materials that are as strong as concrete. For large-scale applications of this material, a quality control system is necessary to minimize defects and avoid wastage. To meet this need, we developed a non-destructive vibration-based sensing system that assesses initial material quality, while still wet, before hardening occurs. We developed a set of artificial intelligence (AI) algorithms that use a set of physics-based features to aid in the smart manufacture of BSC. Our system allows for defect detection early on in the manufacturing process, so that the wet material can be salvaged when defects are identified. Another benefit is that our sensing system enables the monitoring of moisture loss as the material dries out and gains strength. Both of these functions are essential for large-scale deployment of this building material.

The development of biopolymer-bound soil composite (BSC) epitomizes a paradigm shift toward sustainable construction practices that harness the desirable properties of biological polymers to bind granular materials to form composites^{1–3}. BSC is a cement-free building material, as strong as concrete, that is desirable because it has a negative carbon footprint^{1,3–6}. These features of BSC, together with the wide availability of biologically based polymers, serve as a strong rationale for its commercial development. When implemented, BSC will be able to reduce the embodied carbon of the built environment⁴.

Quality control is essential to ensure that materials used to construct the built environment are free of defects, have adequate strength, have consistent mechanical properties, and exhibit long-term durability^{7–9}. Quality control helps to set the initial health of a structure¹⁰. Improperly manufactured materials lead to structural failures, resulting in loss of life. For example, the collapse of the Rana Plaza building in Bangladesh resulted in 1130 deaths. Quality control during material manufacture can play a substantial role in limiting catastrophic failures such as this^{11,12}.

Currently, non-destructive quality control of building materials is performed by a wide assortment of tests, including rebound hammer¹³, penetration resistance¹⁴, pull-out¹⁵, ultrasonic pulse velocity¹³, acoustic emission methods¹⁶, digital image correlation¹⁷, and vision-based methods^{18,19}. Collectively, these tests provide a diverse range of

information regarding the integrity of building materials^{20,21}. While the lack of a quality control method is not an issue for lab development of BSC, it is needed for large-scale manufacture of BSC—for both defect detection and for strength gain monitoring. Examples of defects include voids in the material, internal cracks, and material inhomogeneities, which could lead to material or structural failure. Furthermore, BSC gains strength by solvent loss, which is distinctly different from the mechanism by which concrete gains strength (through a hydration reaction)^{22,23}. Tracking strength gain, however, via mass loss is only feasible for easily handled BSC products (e.g., bricks, tiles, etc.). In the future, BSC products that are too large to be weighed or require manufacture in situ (pavement) may be developed. A non-destructive material assessment technology, capable of both defect detection and monitoring strength gain, that can operate in situ, would be highly desirable for the large-scale manufacture of this biocomposite.

This study introduces a vibration-based sensing system technology (Fig. 1) to enable smart manufacturing of biocomposite materials. Existing non-destructive testing methods primarily focus on hard materials. Our system, however, is designed to perform quality control for biocomposites while they are still soft (in the wet state). Early stage detection of quality issues is important as material in the wet state can be more easily salvaged, thus helping to reduce material waste from the manufacturing process. With our technique, we apply an impulse force (*via* an impulse hammer) onto the

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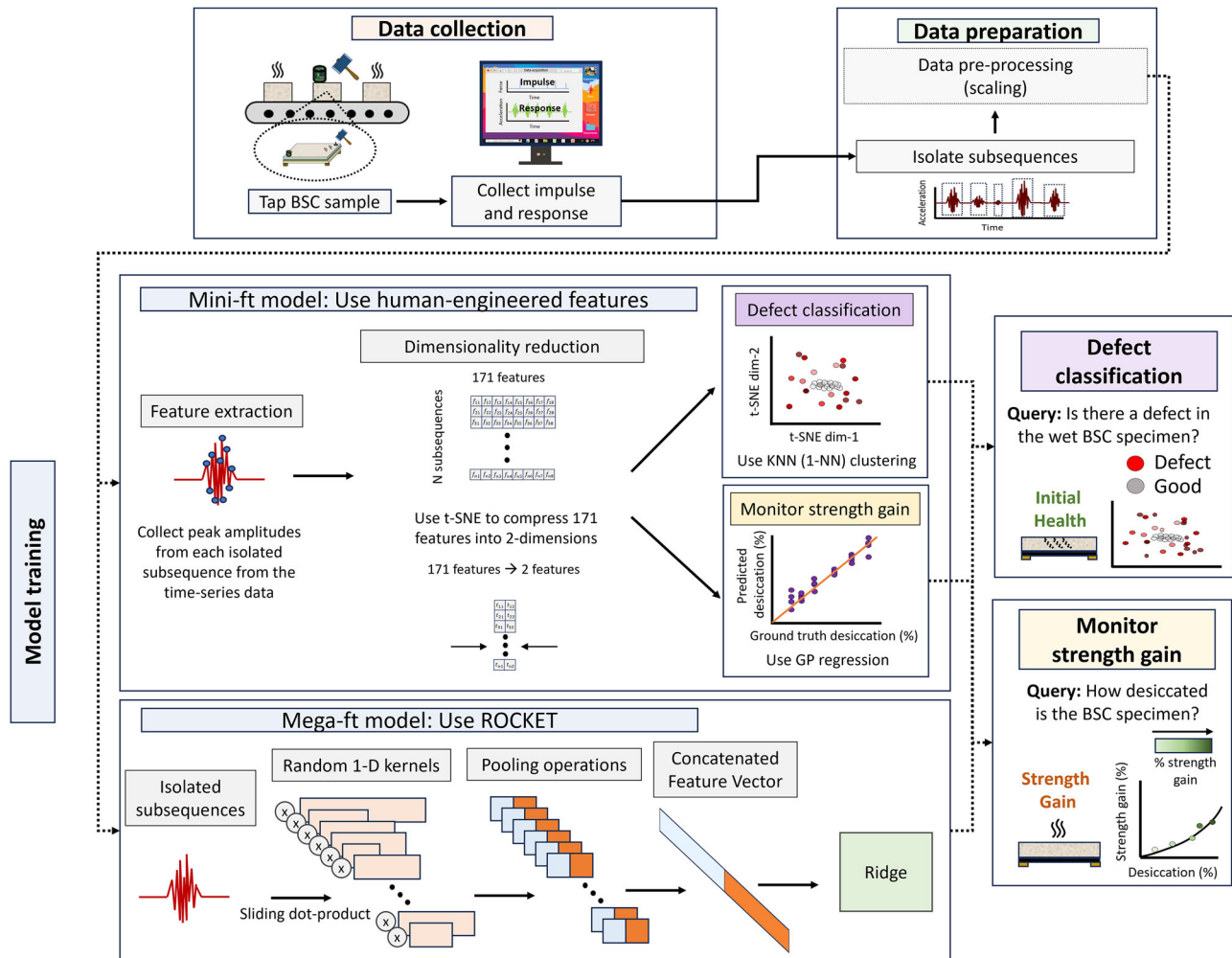


Fig. 1 | AI-powered smart manufacture of biopolymer-bound composites using simple vibration-based testing. Quality control (detection of irregularities and monitoring of strength gain) for smart manufacture of biocomposites can be summarized in four main steps. Our approach uses vibration-based data collected after applying an impulse hammer to a BSC sample. After data collection, the response signal is segmented into subsequences, each linked to a hammer tap. Further processing of these subsequences involves scaling to address impulse strength variations. For Mini-ft, 171 features are extracted, representing the response amplitudes of an isolated subsequence. T-distributed stochastic neighbor embedding (t-SNE) is

used for dimensionality reduction to two features. For defect classification, the K-nearest neighbors (1-NN) algorithm is employed to train models for Mini-ft. For monitoring strength gain, Mini-ft performs Gaussian process (GP) regression using the two t-SNE features as inputs to predict a desiccation level, which can be used to infer the amount of strength gain in a specimen using known relationships for protein-based BSC. A more powerful algorithm, the Mega-ft model, feeds the isolated subsequences into the ROKET classifier, which, while computationally demanding, provides more comprehensive analysis with better accuracy for defect detection and monitoring of moisture loss while the BSC gains strength.

surface of the biocomposite. The resulting vibration waves propagate through the material and can be collected by a vibration sensor (e.g., an accelerometer). Our system is efficient in the following ways: it uses off-the-shelf hardware; the measurement system can be set up quickly; and, it uses efficient AI-powered analytical algorithms, in both training and inferencing, so that rapid analysis can be performed. Furthermore, our approach is dual-functioning in the sense that it is used for both the detection of internal defects (e.g., regions with density anomalies and delamination/internal crack development) and the monitoring of strength gain.

AI-driven systems have increasingly been used to augment traditional approaches to non-destructive testing. For instance, such systems have been applied for real-time monitoring of concrete structures, providing continuous feedback on their structural state²⁴. The incorporation of AI techniques, including artificial neural networks, unsupervised learning, deep learning, recurrent neural networks, and convolutional neural networks, has significantly improved the effectiveness of structural health monitoring and non-destructive testing methods^{25–28}. These advancements have greatly expanded the applicability of non-destructive testing for construction

materials and have facilitated the development of digital twins for civil infrastructure^{29,30}.

Our approach integrates vibration-based sensing with AI models to enable real-time defect detection and strength-gain monitoring. A unique aspect of this technology, lies in its ability to detect defects in building materials at their initial stage of manufacture (i.e., before the material gains strength and remains in a “wet” state, see Supplementary Fig. 1). In short, our system extends previous advances in AI-powered construction material monitoring by offering a robust and scalable approach for quality control and strength-gain monitoring.

Our method is time-efficient and convenient, as the accelerometer can be detached and reattached to different BSC specimens quickly, similar to the way a stethoscope is easily used to check the health of a patient. Additionally, our system is not restricted by a requirement for one particular location for tapping or one particular location for accelerometer placement. To analyze the impulse and response data, we used two different time series-based machine learning pipelines: “Mini-ft,” a simple method that uses human-engineered features (171-dimensional feature space); and,

“Mega-ft,” a more rigorous method that extracts many additional features (20,000-dimensional feature space) for in-depth analysis. Mini-ft and Mega-ft are trained using data from isolated response subsequences after undergoing data preparation (i.e., isolating individual subsequences with the length of 171 data points, followed by data preprocessing such as scaling). Mini-ft collects all of the data points in each isolated subsequence and performs t-SNE to reduce the 171 input features into a two-dimensional feature space. From there, Mini-ft uses KNN (1-NN) for defect classification and Gaussian process (GP) regression for monitoring moisture loss as the material gains strength. On the other hand, Mega-ft takes the data from isolated subsequences after they have undergone data processing and uses the ROCKET classifier³¹, which can be tailored for both classification and regression tasks—defect classification and strength gain monitoring, respectively. See the Methods section for a more detailed description of the machine learning pipelines for both Mini-ft and Mega-ft.

We manufactured a series of BSC specimens, including eight with modeled defects. Our system was able to perform nine-class defect classification (normal vs. one of eight types of modeled defects) with an accuracy of 82% using the Mini-ft model and 97% using the more comprehensive Mega-ft model. Additionally, our method achieved a two-class defect detection (normal vs. some type of defect, unspecified) accuracy of 96% with the Mini-ft model and 99% with the Mega-ft model. Damage quantification, or evaluation of the severity of each defect, can be achieved by utilizing the normalized anomaly scores generated by our models. Finally, our system effectively monitored the strength gain of the composite during desiccation, with a mean absolute percent error (MAPE) as low as 2%.

Results

For this study, 20 rectangular specimens (120 mm $\pm$ 2 mm long $\times$ 90 mm $\pm$ 1 mm wide $\times$ 25 mm $\pm$ 1 mm depth) of protein-based BSC (AP920, a mixture of blood proteins, as the biopolymer) were manufactured according to a single mix design (300 g sand, 50 g water, 50 g AP920). Experiments were carried out using different combinations of sensor location and tap location in order to determine if defect detection was location-dependent. Table 1 shows the types of specimens fabricated together with the amount of data collected. In addition to the specimens listed in Table 1, one specimen was created to collect data for monitoring the material through the desiccation process.

Eleven normal specimens were fabricated. “Normal” in this study refers to BSC samples with no known defects. Eight different samples with various modeled defects were fabricated as shown in Table 1. High-density defects might be present in BSC specimens as a result of mixing inhomogeneities or unusually large aggregate pieces or contaminants. To simulate these high-density defects, we embedded a steel ball ($d = 19$ mm) into one BSC specimen and a steel cylinder ($d = 12.5$ mm and $h = 75$ mm) into another. To detect low-density defects, which might be present as a result of void spaces that might form in a specimen, hollow 3D printed spheres/cylinders (1-mm thick walls) or plastic petri dishes with lids (1-mm thick walls, $d = 40$ mm and $h = 12$ mm) were used. Finally, to simulate specimens with internal cracks or delamination, thin plastic films were placed in the middle of the specimen, with different orientations and sizes (Diag_C: $h = 10$ mm and $w = 20$ mm, Ssquare_C: square 30 mm $\times$ 30 mm, and Bsquare_C: square 70 mm $\times$ 70 mm). The plastic films were ~ 100 microns thick, which is smaller than the maximum allowed flaw size of 150 microns for concrete under different exposure conditions (ACI committee 224, AASHTO LRFD, BS-8110, and Eurocode 2)³². If our system can detect the presence of a piece of plastic film in the specimen, it should be able to detect cracks in the BSC.

Experimental setup and data collection

Figure 2a shows an overview of the manufacturing process of the BSC. The process involves mixing an aggregate (grade 90 silver sand) with a biopolymer (animal blood protein, AP920) together with a solvent (de-ionized water) that dissolves the biopolymer. This material, after mixing, constitutes the “wet mix.” Immediately after the wet mix is created, it is put into a

rectangular mold and compacted to form a BSC specimen. This specimen is immediately subjected to vibration testing, before any desiccation takes place.

Figure 2b shows the experimental setup for collecting data. Wax adhesive was used to affix the BSC specimen to a rectangular platform that was dynamically isolated from the concrete floor by the use of foam dampers. The accelerometer (sampling frequency = 2048 Hz) was fixed to the surface of the specimen by the same wax adhesive. The accelerometer was used to measure the z-axis response at location 1 (bottom left corner) and location 5 (middle of specimen), as shown in Fig. 2c. We used a tri-axial accelerometer (Model No.354C03) from PCB Piezotronics Inc. (Depew, NY). The frequency range ($\pm 5\%$) of the accelerometer is 0.3 to 4000 Hz. The sensor was connected with a digital acquisition system from National Instruments Inc. (Austin, TX) to convert the analog signal to a digital time series. Each impulse force was applied to the specimens by a PCB impact hammer (Model No.086D20). For these experiments, an impulse force using the hammer was applied manually. At each combination of sensor and tap location, data was collected for five 10-s time periods, with ~ 10 –20 taps each time period. Table 1 shows the total number of taps imparted on each type of BSC specimen. Figure 2d, e shows an example of the time series data, and how subsequences can be isolated from the collected data. Figure 2f, g shows how feature extraction occurs in the mini-ft model and the mega-ft model.

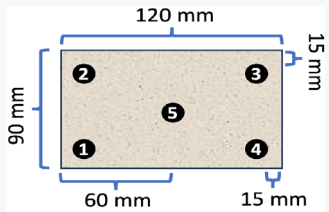
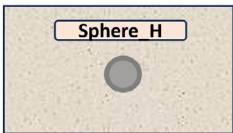
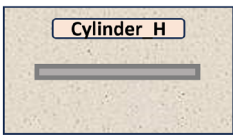
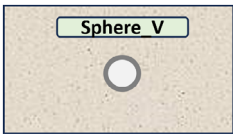
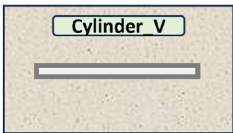
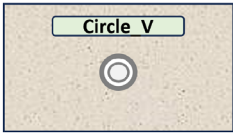
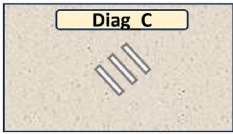
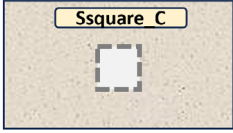
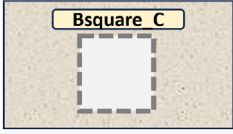
For defect detection, we used a specimen that was freshly manufactured, for which no moisture loss had yet occurred. We investigated the placement of the accelerometer at the corner of the specimen and at the center of the specimen; hence, we collected data only from that location (location 1). After analyzing the initial results, we found that our system performed better when the accelerometer was placed at the corner of the specimen. Hence, we only collected data from location 1. With the position of the accelerometer at location 1, we used the impulse hammer to tap at locations 2, 3, and 5.

For measuring strength gain over the course of desiccation, a normal sample (without defect) was attached to the dynamically isolated platform. An accelerometer was fixed at location 1, with tapping occurring at location 5 for five 10-s time intervals. We examined a single specimen and monitored its mechanical response at various stages of desiccation.

Detection and classification of defects in wet biocomposite

Our mini-ft model was designed to obtain quick results with minimal computational complexity. The inputs to mini-ft consist of 171 evenly spaced time series points of an isolated subsequence corresponding to ~ 16 min-max peaks. Dimensionality reduction using t-SNE was applied to reduce the 171 selected features to a two-dimensional feature space, from which various clustering approaches were explored. The model’s output for defect classification indicates whether the tapped specimen is normal (i.e., without a defect) or defective (i.e., exhibiting one of the specific types of defects). After evaluating several standard classifiers, Mini-ft employs a k-nearest neighbor approach, considering only the nearest neighbor (1-NN), based on the two t-SNE features calculated from the data extracted from the isolated subsequences. To mitigate the influence of variation in impulse strength on the measured response, a scaling approach was implemented, where the response data was normalized according to the strength of the hammer impulse and scaled to a range of -1 to 1 using max-absolute scaling (see Supplementary Fig. 2). If we use mini-ft for defect detection (i.e., to determine “defect vs. no defect”, without specifying the type of defect), the Mini-ft model achieves an accuracy of at least 96%, with or without scaling. For defect classification (i.e., determining if a BSC specimen is normal or contains one of the 8 types of unique defects), Mini-ft achieves an accuracy of at least 78% without data preprocessing (scaling). With data preprocessing, the accuracy of mini-ft for defect classification increases to at least 82%. Figure 3 shows a visualization of the t-SNE features extracted from the isolated subsequences (after preprocessing the data using scaling), which serve as the input for Mini-ft’s k-nearest neighbor (1-NN) classification.

Table 1 | Overview of experimental test specimens and collected data

Top-down view of specimen	Specimen/defect details	Sensor (S) and tap (T) locations
	• Batch 1 (1 specimen) • Batch 2 (5 specimens) • Batch 3 (5 specimens)	B = Batch (e.g., B1 = Batch 1) • B1: 577 [S @ 1 & T @ 2, 3, 5] • B2: 1069 [S @ 1 & T @ 2, 3, 5] • B3: 244 [S @ 1 & T @ 2, 3, 5]
	Specimen Sphere_H contains a solid steel sphere (Diam. = 19 mm).	Total taps = 249 • Sensor @ 1 & tap @ 2 = 85 • Sensor @ 1 & tap @ 3 = 81 • Sensor @ 1 & tap @ 5 = 83
	Specimen Cylinder_H contains a solid steel cylinder (L = 75 mm, Diam. = 12.5 mm).	Total taps = 270 • Sensor @ 1 & tap @ 2 = 90 • Sensor @ 1 & tap @ 3 = 90 • Sensor @ 1 & tap @ 5 = 90
	Specimen Sphere_V contains a hollow sphere (Diam. = 19 mm) made of polylactic acid (PLA).	Total taps = 223 • Sensor @ 1 & tap @ 2 = 73 • Sensor @ 1 & tap @ 3 = 74 • Sensor @ 1 & tap @ 5 = 76
	Cylinder_V contains a hollow cylinder (L = 75 mm, Diam. = 12.5 mm) made of polylactic acid (PLA).	Total taps = 233 • Sensor @ 1 & tap @ 2 = 79 • Sensor @ 1 & tap @ 3 = 78 • Sensor @ 1 & tap @ 5 = 76
	Circle_V consists of a hollow petri dish and lid (H = 12 mm, Diam. = 40 mm).	Total taps = 262 • Sensor @ 1 & tap @ 2 = 86 • Sensor @ 1 & tap @ 3 = 89 • Sensor @ 1 & tap @ 5 = 87
	Diag_C contains three diagonal plastic sheets (H = 10 mm, W = 30 mm, thickness = 60 μ m)	Total taps = 203 • Sensor @ 1 & tap @ 2 = 63 • Sensor @ 1 & tap @ 3 = 71 • Sensor @ 1 & tap @ 5 = 69
	Ssquare_C consists of a square plastic sheet placed horizontally (H = 30 mm, W = 30 mm, thickness = 60 μ m)	Total taps = 241 • Sensor @ 1 & tap @ 2 = 79 • Sensor @ 1 & tap @ 3 = 80 • Sensor @ 1 & tap @ 5 = 82
	Bsquare_C consists of a square plastic sheet placed horizontally (H = 70 mm, W = 70 mm, thickness = 60 μ m)	Total taps = 217 • Sensor @ 1 & tap @ 2 = 74 • Sensor @ 1 & tap @ 3 = 74 • Sensor @ 1 & tap @ 5 = 69

The Mega-ft model is a further enhancement of our system. Similar to Mini-ft, the inputs to Mega-ft consist of isolated response subsequence data obtained from tapping the BSC specimens. The output of Mega-ft for defect classification similarly involves determining whether a BSC specimen is normal or defective. If a defect is present, Mega-ft further identifies the specific type of defect. To

create Mega-ft, we considered 11 different off-the-shelf time series classifiers (see Table 2). As for Mini-ft, we used scaling, which helps to account for variations in impulse hammer force and improves the performance of the model. We selected ROCKET (one of several available time series analysis models), because of its accuracy and computational efficiency.

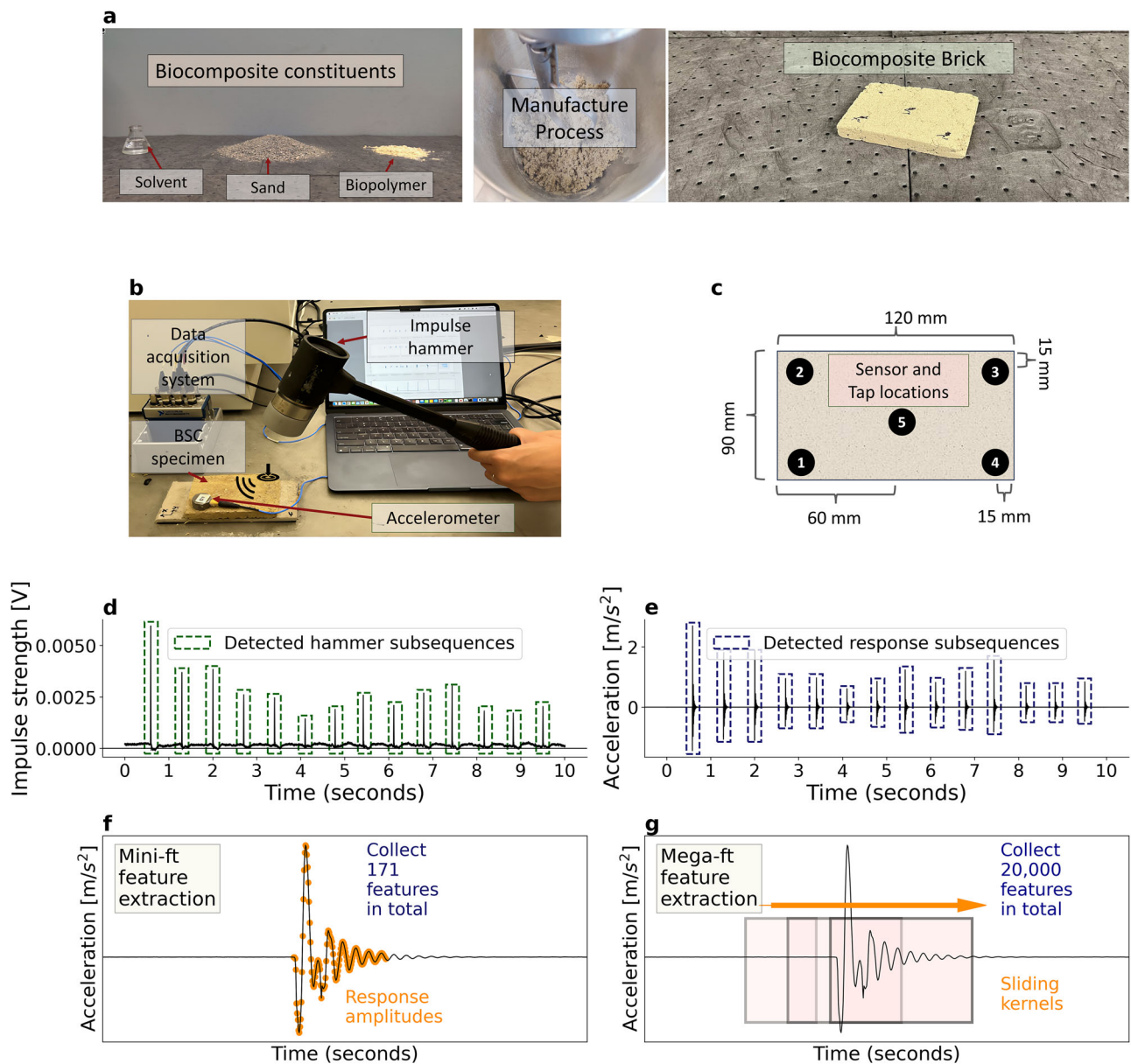


Fig. 2 | Overview of specimen manufacture, data collection, and feature extraction for Mini-ft and Mega-ft. **a** BSC specimens are manufactured by blending the ingredients (aggregate, biopolymer, and solvent) to form a wet mix, which is shaped and compacted into a rectangular specimen. **b** Data collection involves the use of an impulse hammer to tap the BSC specimen, generating vibrations recorded by the accelerometer. The data acquisition system records impulse strength and response time series data. **c** Sensor placement and tapping location on a specimen were varied, with five 10-s time intervals of data being recorded for a given sensor-tapping

location pair. The impulse hammer was tapped manually at a rate of ~1–2 taps per second, with no overlap of response time series data. **d, e** show the recorded impulse and response data, where the dashed boxes show individual vibration response profiles for each hammer tap, which we refer to as subsequences. To account for variations in impulse strength, the response data were normalized w.r.t. the impulse strength, followed by max-absolute scaling. **f, g** show how feature extraction was performed on the subsequences.

Our system was able to effectively determine the initial quality of biocomposites (prior to desiccation), with defect detection accuracies of up to 99% and defect classification accuracies of up to 97% (Table 2). Even with Mini-ft, the accuracy for defect detection (a two-class classification problem, Fig. 4a) was 96 and 99% (Fig. 4b) when we use Mega-ft. Similarly, for defect classification (a nine-class classification problem, Fig. 4c, d), Mini-ft achieved an accuracy of 82%, which can be improved to 97% when Mega-ft is used. Although the accuracy of Mini-ft is notably lower than that achieved using Mega-ft, its performance is still relatively robust. The Mini-ft model can clearly identify normal BSC specimens and BSCs specimens with various types of defects, with the exception of specimens with small cracks or de-bonded defects (i.e., a specimen with a small square crack defect is sometimes mistaken for a specimen with a larger square crack defect). The

Mega-ft model addresses these limitations by considering significantly more features than the Mini-ft model, thus providing the information needed to effectively perform defect classification.

Defect quantification

Defect quantification can be achieved with our system by calculating anomaly scores for each unique type of defect. For Mini-ft (Fig. 5a), a one-class support vector machine (SVM) was used to compute anomaly scores based on the features obtained from dimensionality reduction using t-SNE (t-SNE dimension 1 and t-SNE dimension 2), with a radial basis function kernel and a nu of 0.1³³. Mega-ft uses ROCKAD, which leverages the ROCKET transformation for feature extraction and a k-nearest neighbor approach for defect detection³⁴.

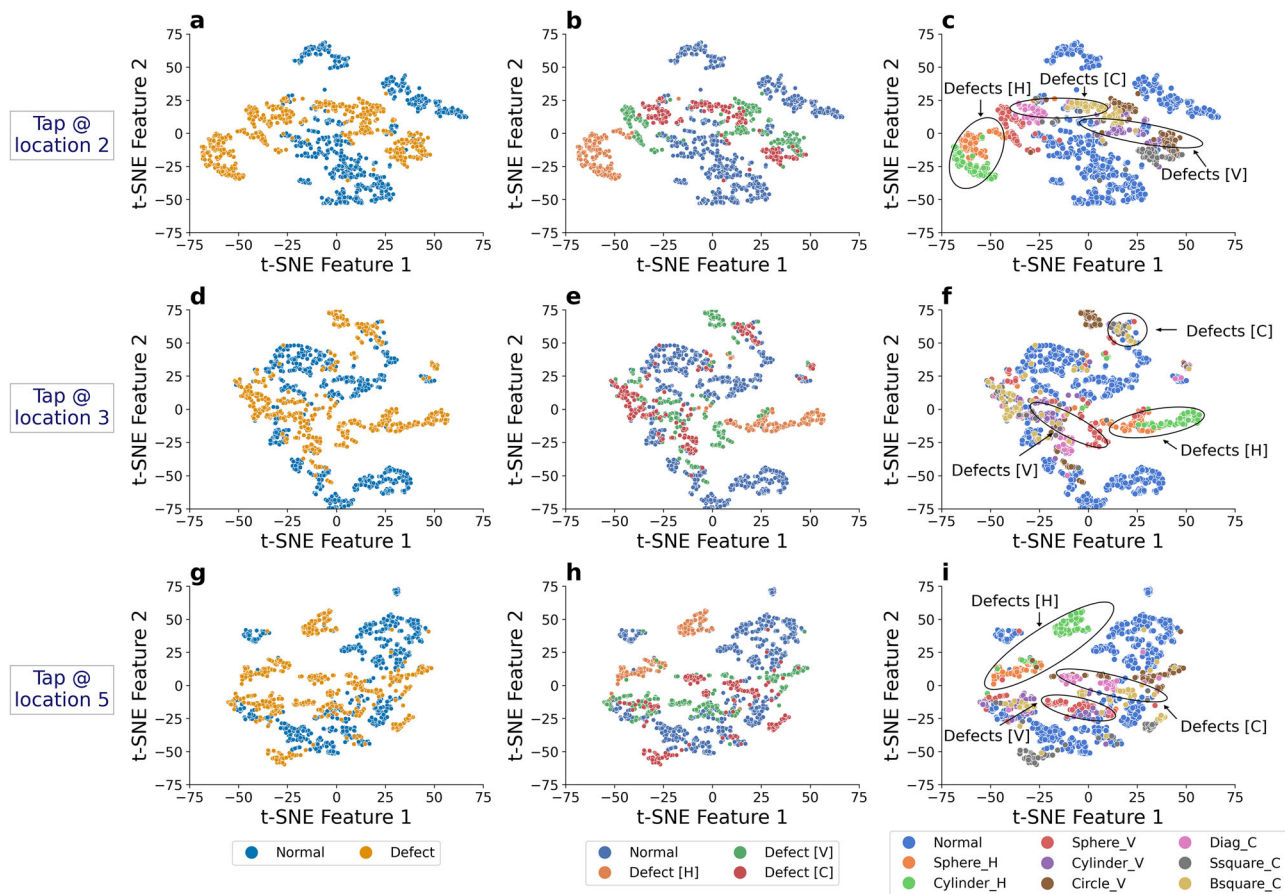


Fig. 3 | Visualization of response features in samples with various defects using Mini-ft for three different tap locations. Dimensionality reduction in Mini-ft was accomplished using t-SNE. **a–c** show the performance of Mini-ft when tapping is conducted at location 2, where panel **a** shows simple defect detection (defect vs. no defect); panel **b** shows defect classification for four classes (normal vs. high-density

defects [H] vs. defects simulating a void space [V] vs. defects simulating a crack [C]); and panel **c** shows defect classification for eight specific types of modeled defects. Similarly, **d–f** show the performance of Mini-ft when tapping is conducted at location 3, and **g–i** show the performance of Mini-ft when tapping is conducted at location 5.

Table 2 | Mega-ft machine learning model candidates for time series classification

Candidate Algorithms	"Normal" vs. "BSC with a defect" (2-class)		"Normal" vs. "8 defect types" (9-class)		
	Defect detection accuracy	F1-score (macro-avg)	Defect classification accuracy	Range of class accuracies	F1-score (macro-avg)
KNN (1-NN)	91%	0.92	76%	54–100%	0.75
TSFresh	90%	0.91	85%	68–100%	0.84
Catch22	83%	0.84	68%	31–90%	0.66
RISE	85%	0.90	88%	74–100%	0.88
HIVECOTEV2	97%	0.98	98%	91–100%	0.98
WEASELV2	98%	0.97	99%	95–100%	0.99
IndivBOSS	80%	0.82	69%	42–90%	0.67
SumClsfier	73%	0.76	54%	33–70%	0.54
ForestClsfier	90%	0.91	81%	72–90%	0.81
ShapeletTrs	91%	0.92	83%	68–100%	0.84
ROCKET ("Mega-ft")	99%	0.99	97%	89–100%	0.97

The ROCKET classifier is used in the final Mega-ft model.

The anomaly scores for Mini-ft were not very distinguishable between normal and defect BSC specimens. However, by considering additional features, the anomaly scores computed using Mega-ft become more distinguishable (Fig. 5b), as specimens with defects modeled using high-density

materials had the high anomaly scores (Fig. 5c). For defects within the same material class, specimens with larger defects had higher anomaly scores than specimens with smaller defects (e.g., the anomaly score of a larger defect, Cylinder_H, is higher than the anomaly score of a smaller defect, Sphere_H).

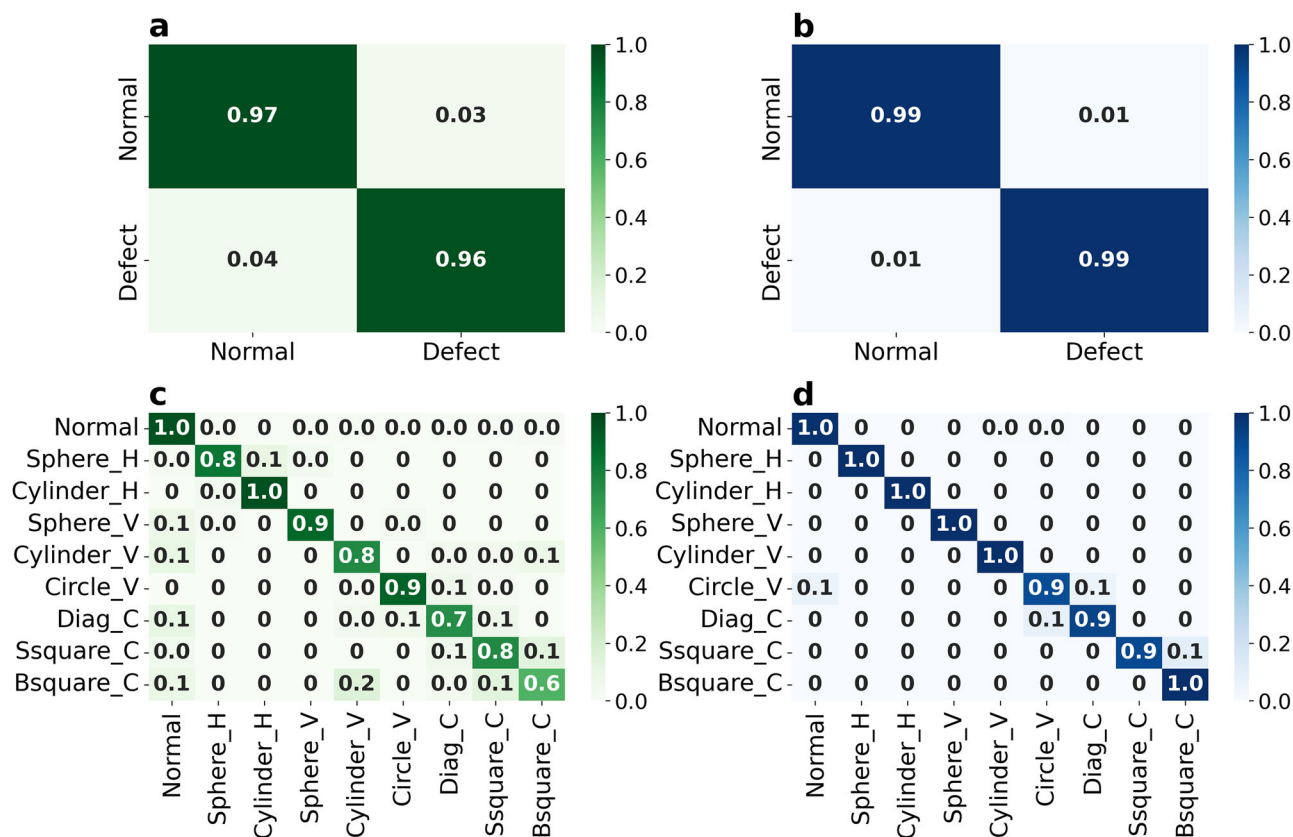


Fig. 4 | Performance of AI-powered analytical algorithms, Mini-ft and Mega-ft, for defect detection and defect classification of BSC. Mini-ft was able to perform defect detection (a) and defect classification (c) with 96% accuracy and 82% accuracy, respectively. Mega-ft performed even better, with an accuracy of 99% for defect detection (b) and an accuracy of 97% for defect classification (d). For defect

classification (nine-class, normal vs. eight unique defects), Mini-ft was able to clearly distinguish between unique types of defects, albeit with minor errors for specimens with microscale defects (Diag_C, Ssquare_C, and Bsquare_C). By considering more features, Mega-ft was able to achieve a higher performance for both defect detection and defect classification.

Monitoring the strength gain of biocomposite

The process of monitoring strength gain is essential in the manufacture of BSC, a solvent-based biocomposite that relies on solvent removal to solidify. In this study, we evaluated specimens of BSC over the course of complete desiccation. Figure 6 shows the results using Mini-ft and Mega-ft for determining the level of desiccation, which can be converted to strength using previously established relationships²³.

The features extracted from the Mini-ft model (the two t-SNE dimensions after performing dimensionality reduction on the response amplitudes of an isolated subsequence) were used to train a Gaussian process regression (GP) model, using an exponential kernel, to estimate level of desiccation. The “ground truth” level of desiccation for BSC can be determined by weighing the specimens and calculating the amount of evaporated solvent¹⁴. Figure 6a shows the predicted degree of desiccation from Mini-ft vs. the ground truth degree of desiccation. Even with the “simple” Mini-ft model, we achieved a low mean absolute percentage error of 2% for desiccation estimation. The effectiveness of Mini-ft in tracking the degree of desiccation directly translates to biocomposite strength. Mini-ft was able to clearly distinguish between different degrees of strength gain (the scatterplot showing the two t-SNE features in the mini-window) and monitor the increase in strength of biocomposite as it solidifies.

Similarly, Fig. 6c, d shows that the Mega-ft model also can be used to monitor the desiccation or gain in strength of BSC. While, the performance of Mega-ft is slightly worse than Mini-ft (5% MAPE vs. 2% MAPE), both analytical algorithms are still fairly robust in monitoring the strength gain of biocomposites as they complete solvent evaporation.

System validation. To determine if our system is distinguishing between different defects, rather than merely distinguishing between different

bricks, we trained the Mega-ft model with data from half of the specimens (half of the normal specimens and half of the specimens made with a modeled defect) and made inferences based on the data from the other half of the specimens not used in training. Upon performing this test, our approach was still able to perform two-class defect detection, albeit with a noticeable reduction in accuracy from 99 to $87.8\% \pm 5.9\%$ (see Supplementary Fig. 3a). This is not surprising, since we used a mixture of different types of specimens, those with high-density defects and those with low-density defects.

We performed an additional analysis in which we grouped defect specimens as follows: specimens with high-density defects (H); specimens with low-density (void) defects (V); and specimens with crack defects (C). For this experiment, the Mega-ft model was trained using half of the normal specimens and one sample from each defect class and tested on the held-out data. Mega-ft was able to achieve $73.4 \pm 9.3\%$ accuracy for this four-class detection (see Supplementary Fig. 3b), which is still fairly effective given that the defects in the trained and test set are somewhat heterogeneous groups (a specimen with a steel cylindrical defect is different from a specimen with a steel spherical defect). A total of 500 train and test runs were executed for each of these tests, with the train and test split performed randomly. These two analyses—using a mixture of different types of specimens and grouping specimens by defect type—gives us confidence that our system is not detecting trivial differences between specimens.

Impact of sensor and tap location on system performance. To determine the influence of tap locations on the performance of the models used in our system, additional experiments (see Supplementary Fig. 4) were conducted for defect detection (two-class classification) and defect classification (nine-class classification) using data collected when

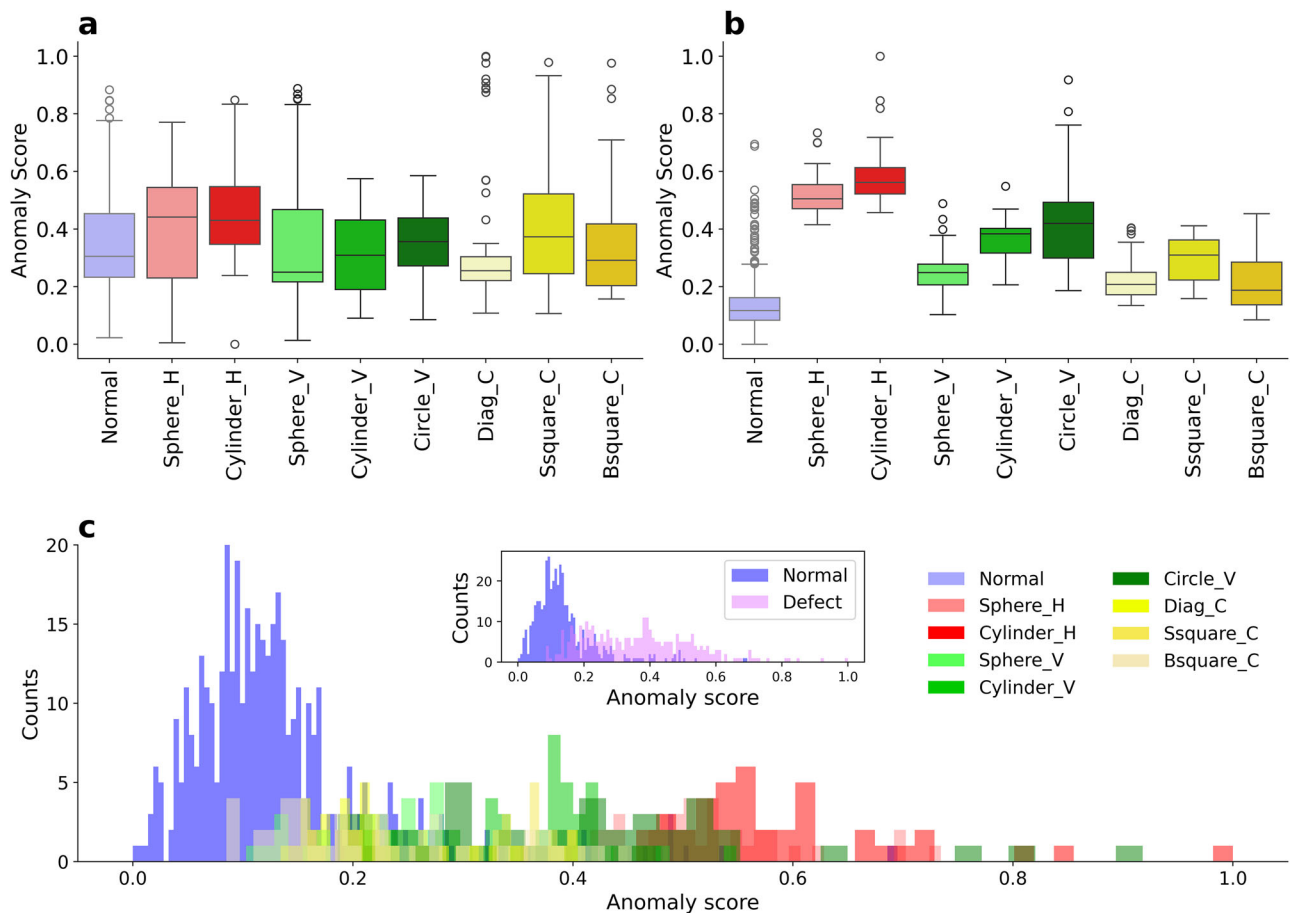


Fig. 5 | Defect quantification in biocomposites inferred through computed anomaly scores for Mini-ft and Mega-ft. Anomaly scores for different types of defects for Mini-ft (a) and Mega-ft (b) were computed. When the anomaly scores of the Mega-ft model are visualized as a histogram (c), it's easy to appreciate the separation and overlap between different types of defects. The highest anomaly

scores from the Mega-ft model correspond to defects modeled with high-density material (red-colored defects with the subscript _H), followed by defects modeled as voids (green-colored defects with the subscript _V), and finally by small crack-type defects (denoted by the subscript _C). Within a given type of defect, specimens with larger defects had higher anomaly scores than specimens with smaller defects.

tapping was performed at location 2 and location 5, with the accelerometer still being placed at location 1.

Our system was able to effectively verify the quality of biocomposites, independent of the tapping location on a given specimen. The Mini-ft model performed best when tapping is conducted at locations 2 or 5. While the accuracy of Mini-ft is lower when tapping is performed at location 3, this decrease in accuracy is not particularly consequential, as the misclassifications made by Mini-ft were minor (i.e., classifying a specimen with a Bsquare_C defect as a specimen with an Ssquare_C defect). Furthermore, our Mega-ft model, which uses many more features, is able to verify the quality of BSC specimens across all tapping locations.

Next, we considered the impact of sensor location on the performance of our system. The accuracy of defect detection dropped only 4% (from 99 to 95%) when the sensor was moved from the corner of the specimen (location 1) to the middle of the specimen (location 5). Since our system can detect defects regardless of sensor placement, we conclude that the performance of our system is largely independent of sensor placement and tapping location.

Implications of the sensing system. Our system addresses a challenge that is inherent in the manufacture of certain biocomposites—namely, the need to detect manufacturing defects in the material while it is in the green (wet) state. The advantage of detecting defects in this early state of manufacture is that the wet material can be returned to a point upstream in the manufacturing process, where it can be re-mixed with the raw ingredients, so as to minimize waste. While there exists a wide range of non-destructive quality control methods, most are tailored for quality

control of hard materials. We have designed our system to perform quality control at the initial state of manufacturing of BSC. Thus, the challenge is to detect flaws in a soft material, which is rarely the arena in which non-destructive testing is applied. An example of a non-destructive technique for evaluation of soft materials is the use of guided elastic wave propagation using optical coherence elastography for characterizing biofilms³⁵. Another example is the use of microwave holographic subsurface technology to detect void spaces in polyurethane foams³⁶. Our study introduces an AI-powered vibration-based sensing system that works with a soft material and is designed for the evaluation of BSC in its wet state. Our technology fills the current gap left by conventional methods. In addition to defect detection, our system is able to perform defect classification, which is beneficial in potentially pinpointing specific issues in the manufacturing process. For example, if we consistently detect specimens with high-density irregularities, we might suspect an issue with the mixing process.

In this paper, we present the results of a pilot study using an AI-powered vibration-based sensing system that represents a notable step toward the smart manufacturing of carbon-negative building materials. To carry out this study, we selected a set of “defect BSC specimen” based on embedding conveniently available objects into the green BSC. Some of these objects are fairly large and may not adequately represent real-world defects that may occur in manufactured BSC products. As a pilot study, however, our investigation was highly successful. We recognize that more realistic defect models, in larger test specimens, would be desirable. By validating the technology on a larger scale, we would be in a stronger position to

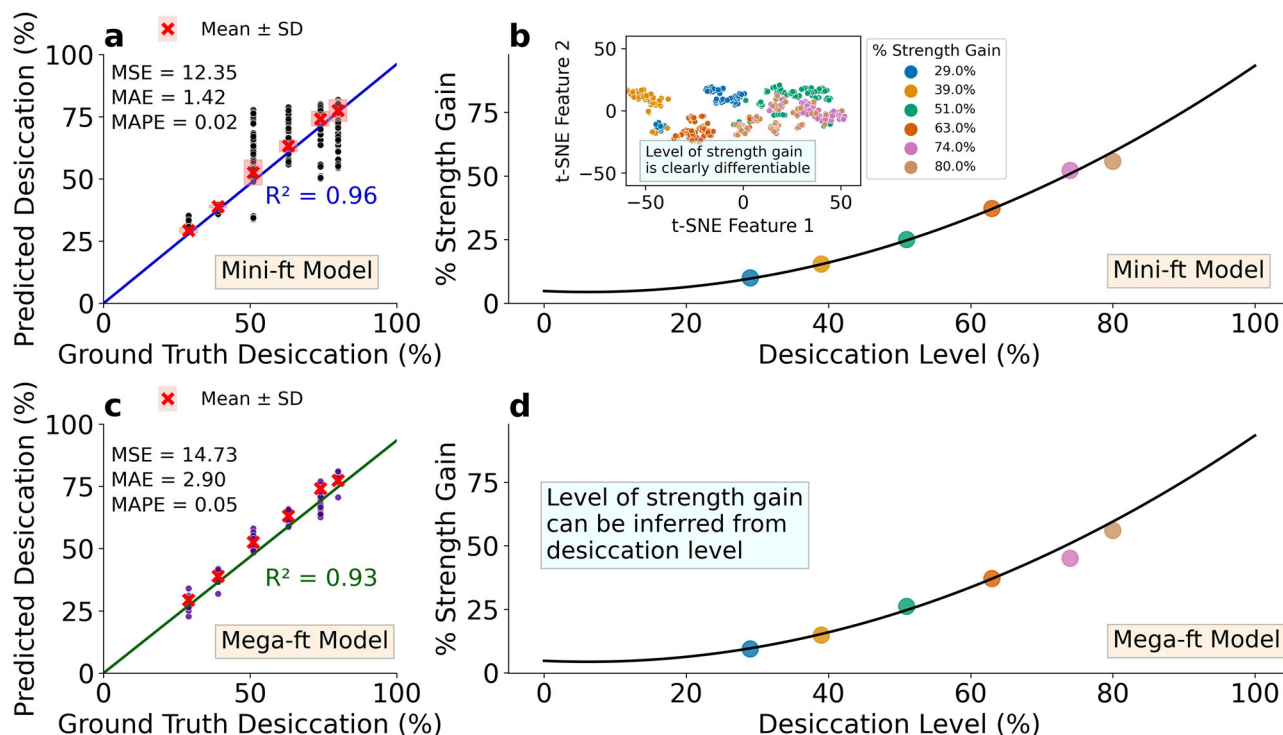


Fig. 6 | Monitoring the strength gain of biocomposites during desiccation using Mini-ft and Mega-ft algorithms. Biopolymer-bound composites transition from an initial wet/soft state to its final hardened state via solvent evaporation. Using the same AI-powered analytical models, we can determine the desiccation level and degree of strength

gain of the material. **a, b** show the performance of Mini-ft in determining the level of desiccation or extent of strength gain of BSC (the black dots correspond to the predicted desiccation according to each isolated subsequence). **c, d** show the performance of Mega-ft for monitoring level of desiccation or strength gain.

demonstrate the utility of our system for in situ pavement applications and larger structural components.

While our system can detect the presence of a defect in the BSC, it cannot identify the location of that defect within the BSC specimen. In a manufacturing setting, it may not be necessary to identify the location of a defect if return of the material to an upstream point in manufacturing is the sole goal. Under some scenarios, determining the location of a defect may be important. This might be accomplished by collecting additional data from different sensor and tap locations. This may be worthwhile for certain types of BSC products that are large and expensive, where salvage of a portion of the product, rather than discarding it completely, makes sense. Further collection of data will help test the limits of vibration-based sensing. One future experiment may involve increasing the separation distance between sensor and tap locations. This may be important for BSC products that are larger or have a layered structure. It is even conceivable that vibration-based testing may be used for quality control of a BSC that is manufactured through a 3D-printing process.

Our study demonstrates the potential utility of non-destructive impact testing for soft biocomposite materials, the size of bricks. With this scenario, it might be reasonable to perform testing on a representative sample of the bricks being produced to ensure that the manufacturing process does not introduce defects. For applications of this technology on larger-sized functional units, such as structural components (e.g., beams) or paving applications, further work will be needed to determine how to adapt this technology for these larger objects.

During the manufacture of BSC building materials, the material design—specifically the composition of the constituents—depends on the requirements of the intended application. These requirements may include desired physical properties (e.g., compressive strength, viscosity of the wet material for 3D-printing applications, and the choice of solvent used for desiccation)². In each of these cases, the proposed system can be effectively deployed for defect detection and strength-gain monitoring. We acknowledge that with different material variants, based on different mix designs, the response signal may show slight

differences in vibration amplitude and frequency. However, as long as the system has been calibrated on a representative set of BSC brick specimens, the system should be able to adapt to these variants.

Our system is also adaptable to varying environmental conditions, such as different temperatures, enabling effective monitoring of strength gain under real-world scenarios. It is well-suited for high-throughput production, as it can be deployed rapidly, with data collection and prediction of defects or strength gain completed within minutes. In practice, during the high-throughput production of BSC construction materials, the sensing system will likely be used to verify the quality of representative samples, particularly between different production batches, to ensure consistent product performance. For strength-gain monitoring during moisture loss, the sensing system can be pre-trained on a range of temperatures at the intended site of application, ensuring the validity of testing results under these conditions.

In the case of environmental variability or sensor noise, signal-processing techniques such as low-pass, band-pass, and Wiener filtering can be employed to mitigate high-frequency electrical and environmental background noise, as well as to address sensor noise and DC offsets in the sensor data^{29,37}. These considerations are especially important in large-scale field deployments of this technology, where testing will be conducted in less controlled environments.

Taken together, the AI-driven system introduced in this study demonstrates how non-destructive sensing modalities can be incorporated into the manufacturing process of sustainable construction materials in an effective and cost-efficient manner, ultimately facilitating the broader adoption of these materials at scale.

Discussion

In this paper, we introduce an AI-driven approach to aid in the smart manufacture of a carbon-negative building material. Our system uses AI-based algorithms that make efficient use of easily acquired vibration data. These analytical algorithms can accurately detect irregularities in a nature-based biocomposite while it is still wet. For most building materials, the

emphasis has been placed on defect detection in the finished material. Our technology takes a different approach in that it allows identification of substandard products at an early stage of manufacture before the material is fully hardened. In fact, our system is even able to monitor the hardening process as the solvent is removed from the biocomposite. The ability to perform material assessment at an early stage is advantageous because early identification of defects allows the raw materials to be salvaged and returned to an upstream stage in the manufacturing process without waste. By contrast, material assessment of precast concrete, which is usually carried out on the finished material, results in rejection of up to 3% of all manufactured products³⁸, without the possibility of capture of the raw material.

The adoption of sustainable materials in the construction industry will likely shift the spectrum of materials away from hard materials toward soft composites of various kinds, similar to the biocomposite material investigated here. We anticipate that the AI-powered sensing technology that we have developed, which is suitable for soft materials, will be widely applicable to many of these construction materials. Both early defect detection and monitoring of solvent removal could enable large-scale deployment of building materials formed of biopolymer-bound soil composite.

Methods

Our AI-powered material assessment system supports the smart manufacture of biocomposites by detecting internal cracks, mix irregularities, and monitoring the strength gain. Unlike quality control systems for conventional materials, which mostly focused on verifying the quality of the finished material^{39,40}, our approach addresses initial material manufacture and strength gain.

Sample manufacturing

To manufacture laboratory-scale specimens of BSC, granular material and biopolymer are mixed for 5 minutes until thoroughly blended. Next, a solvent is added and further mixed for 5 more minutes to form the fresh mix¹. The fresh mix is shaped and subjected to double-sided compaction (5 MPa) using a custom mold and then desiccated in the open or through energy-enhanced methods (i.e., oven) until fully dried out.

A rectangular mold was fabricated by 3D printing using polylactic acid (PLA) as the feedstock. 3D printing was also used to create hollow objects to mimic low-density defects. One of the 3D printed objects was a hollow sphere, and the other was a hollow cylinder, 1-mm wall thickness for each (Sphere_V and Cylinder_V, from Table 1). We also modeled high-density defects by using a steel ball and a steel cylinder. For the final low-density defect, Circle_V (Table 1), a plastic petri dish with a lid was used to create a small cylindrical void in the middle of the specimen.

Data collection and preparation

Vibration data were collected using an impulse hammer to excite various points on the surface of the rectangular BSC specimen, while an accelerometer recorded the corresponding response. To examine the influence of sensor placement and impact location on system performance, the position of both the accelerometer and the tapping points was varied. The specimen was placed on a dynamically isolated platform to minimize background vibrations and ensure that the recorded response reflected only the specimen's behavior. The analog accelerometer signal was converted to a digital signal using a commercial analog-to-digital converter.

Formally, the measurements obtained with our system (see Fig. 6) correspond to a multivariate time series x_T . See Definition (1):

Definition 1. A **time series** $x_T: \vec{T} \mapsto \vec{X}$ is a finite sequence of N data points ordered by time. For each time point t_i in $\vec{T} = \{t_1, \dots, t_N\}$, there is one data point x_i in $\vec{X} = \{x_1, \dots, x_N\}$ in the case of **univariate time series**, and $d > 1$ data points in the case of **multivariate time series**. The time points t_i are used to index the time series, where $x_{T_{t_i \dots t_{i+k}}}$ denotes a **subsequence** s_i .

The multivariate time series x_T consists of two univariate time series: (1) The **hammer time series** x_{T_h} which consists of the pressure measurements obtained from the impulse hammer when it is tapped on the

specimen. (2) The **response time series** x_{T_r} which consists of the vibrations associated with tapping of the hammer measured by the accelerometer.

The machine learning pipeline encompasses two different approaches. The first involves manual feature extraction, which we denote as Mini-ft. A second approach involves the application of time series machine learning models, which we denote as Mega-ft. The time series are first segmented, followed by a preprocessing step (scaling).

Segmentation into subsequences. From the time series x_T , subsequences s_i were extracted such that each subsequence contains one tapping of the hammer and its corresponding vibration response. This was done by identifying and noting the times of each individual peak in the hammer time series data. Since each identified peak is associated with an individual hammer tap, we used the noted times to locate the peaks in the response time series data. Once each peak in the response data was located, a window was defined, from which segmentation of the response data was conducted to obtain subsequences of the response time series associated with each measured hammer impulse (please see Fig. 1). Each subsequence, s_i , is a multivariate time series. Segmentation was achieved by:

- finding the time points t_{peak} of the peaks in the hammer time series.
- for each peak, obtaining a subsequence s_i determined by T_{pre} and T_{post} which specify the number of time points prior and after the detected peak t_{peak} respectively. Hence, $s_i = x_{T_{(t_{\text{peak}} - T_{\text{pre}}) \dots (t_{\text{peak}} + T_{\text{post}})}}$.

A key consideration during the development of our models was determining the appropriate subsequence length for individual responses. During each 10-s data collection interval, the impulse hammer was tapped ~10 to 20 times, producing a recorded response with a corresponding number of distinguishable individual responses. The challenge arose in extracting each response, as incorrect extraction could negatively impact system performance. Defining the start (left-bound of the subsequence, i.e., T_{pre}) of an individual response was not difficult—the start of an impulse was easily identified, as the amplitude of the response peaked early on.

Defining the end (right-bound of the subsequence, T_{post}) was more challenging, as it was difficult to determine when the response signal had fully decayed. We truncated each extracted response subsequence to a total length of 0.08 seconds. This cut-off generally occurred around the 8th min-max peak of a given response, where the amplitude of the peaks on the right-bound had diminished to 5–10% of the maximum amplitude. We explored extending subsequence lengths out to 0.16 and 0.32 s. Extending the length of extracted subsequences did not noticeably affect the performance of the models.

Accounting for hammer impulse variations. Since tapping was conducted manually, variations in the hammer impulse were accounted for by normalizing the response of each subsequence s_i with respect to the hammer impulse. This was achieved as follows:

- for each subsequence s_i determine the maximum value of the hammer subsequence $x_{T_{h_{\text{peak}}}}$
- scale the response subsequence x_{T_r} w.r.t. the hammer impulse using $x' := \frac{x_{T_r}}{x_{T_{h_{\text{peak}}}}}$

Max-absolute scaling. Since some candidate models are scale-dependent, the normalized data is scaled using max-absolute scaling as follows:

- determine the global maximum absolute value $\max(\text{abs}(x))$ of the training set
- scale each input time series using $x'' := \frac{x'}{\max(\text{abs}(x))}$

This scales the training data to a value range constrained by $-1 \dots 1$ without shifting the center (i.e., positive/negative values stay positive/negative). The same scaling parameters are applied to the test data set.

Model selection and training

Upon collecting the data shown in Table 1, we can now develop the Mini-ft model (considers a 171-dimension feature space) and the Mega-ft model

(considers a 20,000-dimension feature space) for verifying the quality of biocomposites while they are still in the wet state and for monitoring the hardening of these materials as they desiccate. By considering two different analytical algorithms—a model using human-engineered features (Mini-ft) and a model using a more rigorous Rocket classifier (Mega-ft)—we can increase the flexibility and versatility of the system (albeit with a trade-off between accuracy and computational costs). Figure 1 shows the AI-powered analytical models developed in our study.

The physics-based features inferred from the time series signals are directly linked to the underlying physical behavior of the BSC specimens. For both the Mini-ft and Mega-ft models, the peak amplitudes for the response signals show the greatest correlation with the stiffness of the BSC specimen. This is particularly relevant for strength-gain monitoring, as the sensing system captures changes in peak amplitudes and relates them to the material becoming stiffer over time. The Mega-ft model incorporates a broader set of physics-based features, enabling it to account for shifts in response signal frequencies as well as attenuation and damping of the material. By leveraging this more extensive feature set, the Mega-ft model can potentially provide a more holistic assessment of the physical state of BSC specimens.

Mini-ft model development. We developed the Mini-ft model to perform quick quality control. This model involves extracting the response amplitudes from each isolated subsequence, resulting in 171 features being extracted (shown in Fig. 2). These response-amplitude-based features are extracted based on the physical insight that signal peaks typically correlate with the mass and stiffness of the material. These features are further compressed into two dimensions through a Euclidean t-SNE operation, to get the two features (first and second t-SNE dimensions) to train the Mini-ft model. Machine learning classifiers for tabular data (i.e., feature spaces) are evaluated: e.g., support vector machines with different kernels, nearest neighbors, decision trees, and neural networks. The k-nearest neighbors (kNN) classifier was applied using the two-dimensional t-SNE features (first and second t-SNE dimensions) as input predictors. Each data sample is represented as a point in this 2D feature space, and classification is performed by comparing the position of a new sample to those of the labeled training samples. The Euclidean distance metric was used to measure similarity between points, with the number of neighbors set to $k = 1$. The features were standardized before training. The t-SNE features used for testing were therefore directly compared with the distribution of the training samples in this 2D embedding space, allowing the kNN classifier to assign class labels based on spatial proximity in the t-SNE representation.

To tailor Mini-ft towards strength gain monitoring, we used the same two-dimensional t-SNE features to train a Gaussian process regression (GP) model to monitor the strength gain of BSC as it undergoes desiccation. Gaussian process (GP) regression is a non-parametric Bayesian approach used for regression tasks. In this framework, we assume that the underlying function can be modeled as a Gaussian process, characterized by a mean function and a covariance function (kernel)⁴¹.

To compute the ground-truth strength gain of the biocomposite material, Equation (1) was used to relate the degree of desiccation of BSC (using protein-based biopolymer binder AP920) during drying to its corresponding level of strength gain²³. This relationship enables the quantification of the degree to which moisture loss over time translates into mechanical strengthening, providing a basis for validating the predictions made by the AI-driven sensing system. Equation (1) describes this relationship:

$$\text{strength (\%)} = 0.0101 * \text{desiccation level}^2 - 0.124 * \text{desiccation level} + 4.756 \quad (1)$$

where, desiccation level refers to the level of desiccation (%), and strength (%) shows the percentage of strength gain in relation to the level of desiccation.

Mega-ft model development. While the Mini-ft model described above is fast and simple, it has two inherent limitations: (1) the model requires manual feature extraction—so-called human feature engineering and (2) the temporal aspect of the data is only considered to a limited extent. Hence, as an alternative, we tested a wide range of machine learning models (classifiers) that work directly on the time series data and do not require manual feature engineering. We evaluated models with different fundamental concepts. The results are shown in Table 2. For all of the models, their implementation provided by *aeon*, a Python toolkit⁴² was used:

- KNN (1-NN): nearest neighbors-based method that classifies based on the distances between the time series segments using Euclidean distance
- TSFresh⁴³ and Catch22⁴⁴: feature-based methods that extract a number of predefined features and classify on that feature space
- RISE⁴⁵: ensemble-based method that combines different base classifiers trained on random intervals
- HIVECOTEV2⁴⁶: an heterogeneous ensemble which we used with a maximum training time budget of 2 min
- WEASEL 2.0⁴⁷: extracts patterns from time series and classifies based on the existence and frequencies of these patterns
- Individual BOSS⁴⁸: extracts a dictionary of patterns (bag-of-words)
- Summary classifier: extracts a predefined set of statistical features (see ref. 42 for details)
- Time series forest⁴⁹: extracts random intervals and trains an ensemble of decision trees on these
- Shapelet transform: a representative from the family of shapelet-based methods based on the ideas in ref. 50, which extracts characteristic shapes from the time series to distinguish between the classes
- ROCKET³¹: a kernel-based classifier exploiting the ideas of kernels (windows) moved over the time series, similar to a one-dimensional convolutional neural network, but using randomized kernels

In our final Mega-ft model, we use the time series classifier ROCKET, proposed in ref. 31. ROCKET generates K convolutional kernels and moves these over the input time series. These kernels correspond to windows with the kernel parameters of weights, bias, dilation and padding. Rather than learning these parameters from the input data, ROCKET uses randomized kernels, i.e., draws the parameter values for each kernel from predefined ranges or distributions. Sliding the kernels over the time series and calculating the dot product of kernel and time series at each position, two features are extracted for each of the K kernels: (1) the greatest value of the dot product and (2) the proportion of dot products that are positive (see original ROCKET paper³¹ for details). We use the default setting of $K = 10,000$ kernels; hence, we obtain a feature space of $2K = 20,000$ dimensions. A linear classifier (Ridge classifier) is trained on the ROCKET-induced feature space. Due to not learning the kernel parameters and due to using a simple classifier, this method is computationally very efficient. We found that the ROCKET classifier has the capacity to capture more intricate vibrational features, such as damping, wave dispersion, attenuation, and interference during vibration wave propagation, which are difficult to capture by signal peaks only (features in Mini-ft).

Evaluation metrics. To evaluate the performance of Mini-ft and Mega-ft in defect detection and defect classification, we reported the accuracy of the models and computed the macro-averaged F1-score (as shown in Table 2). In this case, the macro-averaged F1 scores are obtained by averaging the F1-score for each class, i , across all classes, N :

$$F1_i = 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (2)$$

$$F1_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N F1_i \quad (3)$$

Where $\text{Precision}_i = TP_i / (TP_i + FP_i)$ and $\text{Recall}_i = TP_i / (TP_i + FN_i)$.

To evaluate the performance of Mini-ft and Mega-ft in strength gain monitoring, we computed the mean absolute percentage error (MAPE, see (4)), mean absolute error (MAE, see (5)), and mean squared error (MSE, see (6)) as follows:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (4)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6)$$

Where n represents the number of observations (subsequences), y_i represents the ground truth degree of desiccation, and $\hat{y}_i$ is the predicted degree of desiccation.

Data availability

Data from the experimental tests of the BSC specimens are available at <https://github.com/ml-and-vis/BSC-quality-control.git>.

Code availability

Code for the Mini-ft and Mega-ft models for defect detection and strength gain monitoring, are available at <https://github.com/ml-and-vis/BSC-quality-control.git>.

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Author contributions

B.H.M. and Y.D. defined the problem. A.C.L. designed and 3D printed the modeled defects and molds. B.H.M. and A.C.L. manufactured the biocomposite specimens used in this study. B.H.M., Y.D., and A.C.L. collected the vibration-based data. B.H.M. and A.T. developed the analytical algorithms. B.H.M. and A.T. implemented the algorithms. All authors jointly revised the paper. M.D.L. and D.J.L. jointly supervised the project. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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